

PROBLEM SET 1

1. Linear Equations and Labor Markets

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The Excel file `dataset.xlsx` contains three labor market variables for Mexico, all constructed using the ENOE survey: the unemployment rate $urate_t$, i.e., unemployed workers divided by the labor force; the employment-to-population ratio $epop_t$, i.e., employed workers divided by the working-age population; and the out-of-labor-force ratio $olfrate_t$, i.e., people out of the labor force divided by the working-age population.

(1) Plots of labor market stocks.

Unemployment rate.

As shown in Figure 1, the unemployment rate in Mexico fluctuates around relatively moderate levels during the 2005-2008 time span, followed by a noticeable increase around 2009–2010, coinciding with the global financial crisis. After this episode, unemployment gradually declines over the subsequent years, reaching comparatively low levels by the late 2010s. A sharp spike is observed in 2020, associated with the COVID-19 pandemic and the widespread shutdown of economic activity. Although unemployment decreases afterward, it remains above its pre-pandemic minimum, indicating an incomplete labor market recovery by the end of the sample.

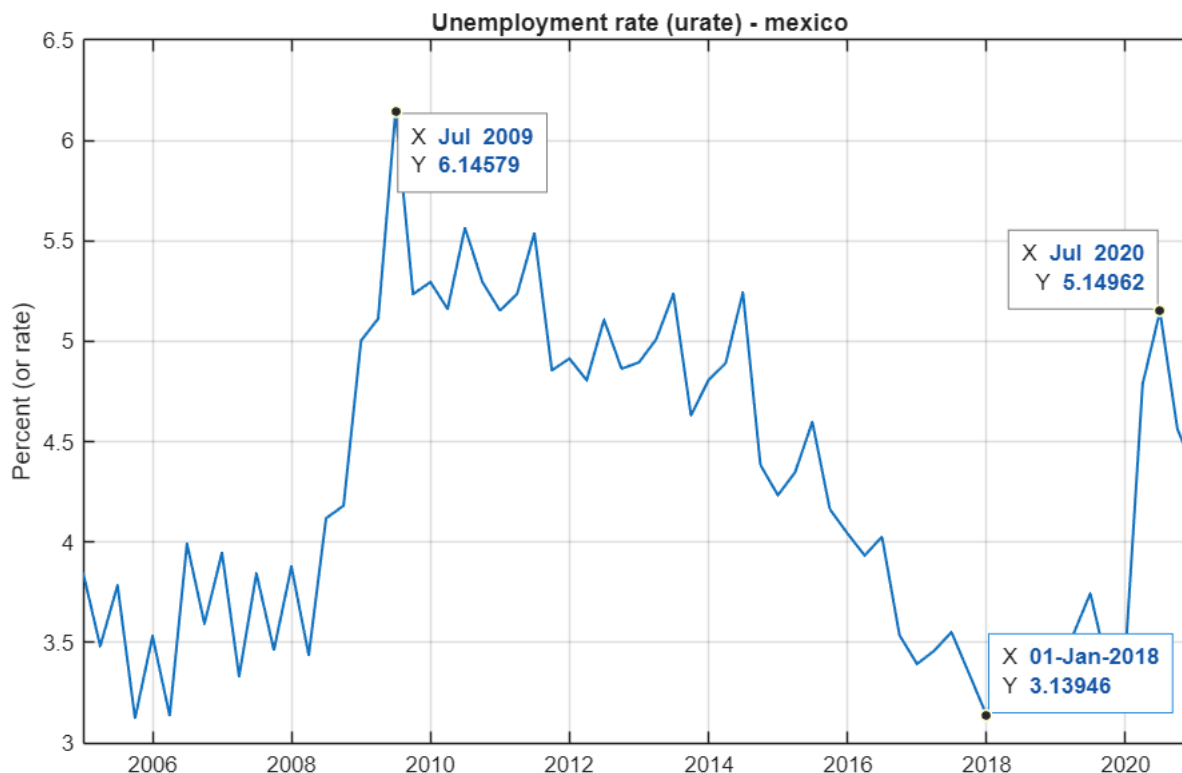


Figure 1: Unemployment rate (Mexico, 2005-2021)

Employment to population ratio.

As it can be seen in Figure 2, the employment-to-population ratio in Mexico remains relatively stable over most of the sample, fluctuating within a narrow range. A mild decline is visible around the time of the global financial crisis (2009–2010), followed by a gradual recovery. The most striking feature is the large drop in 2020, reflecting the massive employment losses generated by the COVID-19 pandemic. Afterward, the ratio rebounds, but it does not immediately return to its pre-pandemic level, suggesting persistent scarring effects in employment.



Figure 2: Employment to population ratio (Mexico, 2005-2021)

Out of labor force ratio.

Figure 3 shows that the out-of-labor-force ratio in Mexico is fairly stable prior to 2020, oscillating around a constant level with no strong long-run trend. In 2020, however, there is a pronounced increase, indicating that a large fraction of individuals exited the labor force during the pandemic, either temporarily or permanently. Although the ratio declines somewhat after 2020, it remains elevated relative to pre-pandemic values, suggesting that a significant part of the employment collapse was absorbed through reduced labor force participation rather than solely through higher unemployment.

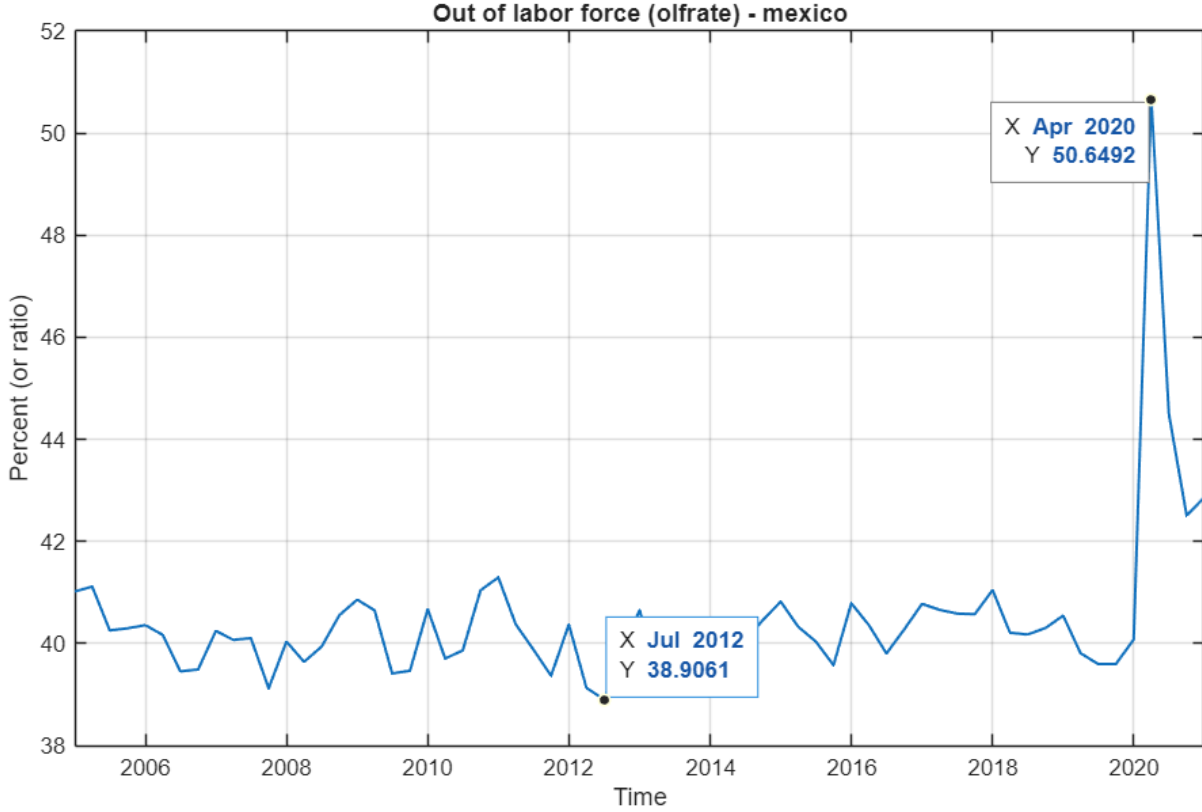


Figure 3: Out of labor force ratio (Mexico, 2005-2021)

(2) Log-log linear regression for Mexico.

We start from the following accounting identity for the employment rate:

$$1 - \frac{U}{LF} \equiv \frac{E/P}{1 - OLF/P}, \quad (1)$$

or, equivalently,

$$1 - \text{urate} \equiv \frac{\text{epop}}{1 - \text{olfrate}}, \quad (2)$$

where P denotes the working-age population, urate is the unemployment rate, epop is the employment-to-population ratio, and olfrate is the out-of-labor-force ratio.

Taking logarithms and applying a first-order Taylor approximation yields

$$d \log(epop) \approx -\frac{ur\bar{ate}}{1 - ur\bar{ate}} d \log(urate) - \frac{olf\bar{rate}}{1 - olf\bar{rate}} d \log(olf\bar{rate}), \quad (3)$$

where $\bar{x}$ denotes the steady-state value of variable x .

This approximation motivates the following log-log regression model:

$$\log(epop_t) = -a \log(urate_t) - b \log(olf\bar{rate}_t) + \varepsilon_t, \quad (4)$$

where a and b can be interpreted as elasticities measuring the contribution of the unemployment rate and the out-of-labor-force rate, respectively, to fluctuations in the employment-to-population ratio, and ε_t is an error term. Adding a constant β_0 , the regression model can be written as

$$y_t = \beta_0 + \beta_1 x_{1t} + \beta_2 x_{2t} + \varepsilon_t, \quad (5)$$

where $y_t = \log(epop_t)$, $x_{1t} = \log(urate_t)$, $x_{2t} = \log(olf\bar{rate}_t)$, $\beta_1 = -a$ and $\beta_2 = -b$.

Using MATLAB, the following results are obtained for Mexico:

Table 1: OLS estimates for Mexico (2005-2021)

Coefficient	Estimate	Standard Error
β_0	7.0038	0.0319
β_1	-0.0460	0.0017
β_2	-0.7826	0.0086

The estimates for the coefficient vector $\hat{\beta}$ are obtained by solving the linear system

$$(X'X)\hat{\beta} = X'y.$$

using Gauss-Jordan elimination.

Standard errors are computed as

$$se(\hat{\beta}_j) = \sqrt{\hat{\sigma}^2 c_{jj}},$$

where c_{jj} denotes the j, j -th element of the matrix $(X'X)^{-1}$.

The estimates of a and b are simply the negatives of the estimates of β_1 and β_2 , respectively,

i.e: $\hat{a} = 0.0460$ and $\hat{b} = 0.7826$. The standard errors of $\hat{a}$ and $\hat{b}$ are identical to those of $\hat{\beta}_1$ and $\hat{\beta}_2$.

(3) Log-log linear regression for Chile (and comparison).

The first point to note is that the data for Chile differ from those for Mexico. While the Mexican data are observed at a quarterly frequency, the Chilean data are observed monthly. In addition, the Mexican sample covers the period 2005–2021, whereas the Chilean sample spans 2010–2021. The same model is estimated for both countries,

$$\log(epop_t) = -a \log(urate_t) - b \log(olfrate_t) + \varepsilon_t, \quad (6)$$

and therefore the interpretation of the parameters a and b is the same in both cases, namely as elasticities. However, the estimation of the coefficients may differ across countries because the Chilean data contain a larger number of observations and capture higher-frequency fluctuations, which can affect the magnitude and precision of the estimated parameters.

Using MATLAB (following the same procedure as before), the following results are obtained for Chile:

Table 2: OLS estimates for Chile (2010-2021)

Coefficient	Estimate	Standard Error
β_0	6.9164	0.0322
β_1	-0.0871	0.0033
β_2	-0.7403	0.1020

Hence, the estimate values for a and b are $\hat{a} = 0.0871$ and $\hat{b} = 0.7403$.

In order to compare the coefficients over the same time period, the model is re-estimated for Mexico using data from 2010–2021. The following results are obtained:

Table 3: OLS estimates for Mexico (2010-2021)

Coefficient	Estimate	Standard Error
β_0	7.0306	0.0359
β_1	-0.0464	0.0023
β_2	-0.7896	0.0097

Note that the estimated coefficients from this Mexico (2010–2021) specification are very similar to those obtained from the previous model.

Comparison.

Comparing elasticities, Chile exhibits a larger unemployment elasticity in absolute value ($a_{CL} = 0.0871$) than Mexico ($a_{MX} = 0.0464$). In fact, the unemployment elasticity in Mexico is roughly one half of that in Chile. This indicates that, in percentage terms, changes in the unemployment rate are associated with much larger movements in the employment-to-population ratio in Chile than in Mexico.

In contrast, the out-of-labor-force elasticity is large and similar in both countries, with Mexico being slightly larger ($b_{MX} = 0.7896$ vs. $b_{CL} = 0.7403$). Overall, fluctuations in the employment-to-population ratio appear to be driven primarily by changes in labor force participation rather than by changes in unemployment. An intuitive interpretation is that in Mexico a substantial part of labor market adjustment occurs through movements into and out of the labor force, whereas in Chile a relatively larger share of adjustment occurs through open unemployment. Consequently, participation margins dominate employment fluctuations in both countries, but unemployment plays a more prominent role in Chile than in Mexico.

Intuitively, the central idea is that, in both countries, declines in the employment-to-population ratio are driven mainly by people leaving the labor force, rather than by increases in open unemployment. In simple terms, many individuals stop looking for work instead of remaining unemployed and actively searching.

There is, however, an important difference between the two countries. In Chile, when economic conditions deteriorate, a larger share of individuals becomes unemployed. In Mexico, many individuals instead exit the labor force altogether and stop searching for jobs. As a result, unemployment plays a more important role in explaining employment fluctuations in Chile, whereas exits from the labor force play a more important role in Mexico.